

AI Assistant Capabilities Overview

A practical guide to the task types this assistant can complete — from structured documents, spreadsheets, and slide decks to charts, data analysis, web applications, and AI-powered media — with concrete examples of what to request and what you receive.

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PERSONAL REFERENCE EDITION · SEPTEMBER 2026

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1. Working With This Assistant

This handbook is a personal reference guide to the practical work this assistant can take on for you. It is organized around task types rather than technical features, because that is how work actually arrives: you need a document, a chart, a cleaned dataset, a slide deck, or a working web page. Each chapter describes one capability area, explains what the finished deliverable looks like, and gives concrete examples of requests that produce it. Keep it nearby as a cheat sheet: when a task comes up, skim the relevant chapter and adapt one of the example phrasings.

Across all areas, the working pattern is the same. First, the request is clarified when key details are missing - audience, length, tone, format, and must-include facts are confirmed before production begins. Second, a short plan or outline is set so you can course-correct early, before effort is invested in the wrong direction. Third, the deliverable is produced as a real file rather than as loose chat text whenever the output is a document, workbook, deck, image, or dataset. Finally, the result passes through automated quality checks - layout validation, font checks, table overflow detection, and table-of-contents verification - before it is handed over.

The table below maps the eight capability areas to their typical deliverables. Chapters 2 through 10 cover each area in depth, Chapter 11 explains how to phrase requests for the best results, and Chapter 12 collects workflow tips, best practices, and answers to frequently asked questions.

Area	Typical deliverables	Example request
Documents & reports	Word and PDF files: reports, proposals, contracts, resumes, exam papers	Write a 6-page market analysis on city e-scooters, formal tone, with charts.
Spreadsheets & data	Excel workbooks with live formulas, summaries, and embedded charts	Build a 2026 household budget in Excel with monthly tabs and a summary.
Presentations	Slide decks (PPTX or HTML) with covers, sections, and speaker notes	Make a 10-slide investor deck about our logistics startup, business style.
Charts & diagrams	PNG charts, flowcharts, mind maps, architecture diagrams, Mermaid code	Draw a flowchart of our customer onboarding process and export it as PNG.
Data processing	Cleaned datasets, analysis summaries, reusable conversion scripts	Analyze this CSV of orders, remove duplicates, summarize revenue by region.
Web applications	Interactive Next.js apps: dashboards, internal tools, landing pages	Build a dashboard that tracks my team's monthly KPIs with editable targets.
AI media	Generated images, edited images, voice audio, transcripts, video analysis	Generate a cover illustration for my blog post about deep-sea exploration.
Research & files	Sourced summaries, extracted tables, merged or converted documents	Find 2026 EV sales figures, then merge the three source reports into one PDF.

Table: Capability areas at a glance

Deliverable-first principle

If the output can be a file - a PDF, workbook, deck, image, or dataset - it is delivered as a file, not as copy-paste chat text.

2. Documents & Reports

Document production is the broadest capability area. It covers structured writing delivered as Word or PDF files: analysis reports, white papers, business proposals, project plans, feasibility studies, contracts, formal letters, resumes and cover letters, exam papers and worksheets, invoices and receipts, and technical documentation. A typical deliverable is a complete, formatted file rather than a rough draft, with a designed cover page, an auto-generated and clickable table of contents, numbered chapters, styled headings, data tables, and embedded charts wherever they help the reader.

Quality standards are built in rather than bolted on. Content follows minimum-depth rules, so sections contain substantive, developed paragraphs instead of thin bullet lists. Layout rules prevent the classic

failure modes: tables never exceed the page width, headings are never orphaned at the bottom of a page, page breaks occur only at structural boundaries, and body text uses consistent professional typography with proper line spacing. Word documents can additionally include tracked changes and comments, which is useful when you need to revise an existing file rather than create a new one from scratch.

When you request a document, the more context you provide, the closer the first draft lands. The most useful details are the intended audience, the target length, the tone (formal, persuasive, academic, or plain), any required sections, and the facts, figures, or sources that must be included. Chapter 11 shows before-and-after examples of vague requests versus precise ones, and the difference in outcomes is usually significant.

Example request	What you receive
Draft a 5-page project proposal for an internal mentorship program, formal tone, for HR leadership.	A PDF or Word file with a cover, table of contents, numbered sections, a budget table, and a rollout timeline.
Turn these meeting notes into a one-page decision memo.	A compact formatted memo highlighting decisions, owners, and deadlines.
Create a 20-question multiple-choice quiz on basic statistics with an answer key.	An exam-style PDF with indented options, section headers, point values, and a separate answer key.
Rewrite my resume for a data analyst role, ATS-friendly.	A cleanly structured single-column resume, with a matching cover letter on request.

Table: Document requests and resulting deliverables

3. Spreadsheets & Data Tables

Spreadsheet work turns raw numbers into organized, self-updating workbooks. You can ask for a new .xlsx file built from scratch - a budget, an inventory tracker, a schedule, a grade book - or for an existing file to be cleaned, reorganized, or extended. Workbooks can contain multiple sheets, styled header rows, frozen panes, sensible column widths, computed columns with real formulas (SUM, AVERAGE, IF, XLOOKUP, COUNTIFS), summary blocks, and charts embedded directly on the sheet. The result behaves like a spreadsheet a colleague built for you: edit an input cell and every dependent total updates.

This area also covers the unglamorous but high-value jobs: converting between CSV, TSV, and Excel formats; splitting one wide table into normalized sheets; merging monthly exports into a single annual file; deduplicating rows; standardizing inconsistent date and phone formats; and flagging outliers with conditional formatting. If you provide messy data, it helps to say what clean means for you - which

columns matter, how duplicates should be resolved, and what the final shape should be. With that, the workbook arrives already in that shape, together with a short log of what was changed.

Example request	What you receive
Build a monthly budget spreadsheet with categories, formulas, and a yearly summary tab.	A multi-sheet .xlsx with live formulas; editing one cell updates all totals.
Take this CSV of sales rows and produce a summary by product and quarter.	An Excel file with raw data on one sheet and a computed summary table plus a bar chart on another.
Clean this contact list: remove duplicates, standardize phone formats, split full names.	A tidy workbook plus a change log listing how many rows were affected and why.

Table: Spreadsheet requests and resulting deliverables

4. Presentations & Slide Decks

Slide deck production starts with structure, not decoration. A deck request produces a planned narrative first - a hook, context, three or four core sections, and a closing - and then renders that structure with consistent visual design. Decks are delivered as .pptx files or as HTML slide documents, with a designed cover, section divider slides, content layouts that vary between bullet slides, two-column comparisons, tables, and image slides, and a deliberate closing slide rather than an abrupt stop.

Two design decisions shape the result most: the visual style and the information density. Styles range from clean business (navy and slate, serif headings) through tech and futuristic to education and warm; pick one and it is applied consistently across every slide, including fonts, accent colors, and corner treatments. Density is likewise a choice: sparse keynote-style slides that carry one idea each, or denser briefing decks that read well without a presenter. Speaker notes can be included per slide, either as brief cue cards of one or two sentences or as a full verbatim script you can read aloud.

As a calibration rule, plan on roughly one to two minutes of speaking time per content slide; a ten-minute talk therefore lands at about eight to twelve slides plus cover and closing. Two example requests: Make a 10-slide deck introducing our Q3 results to the board, clean business style, with a one-line takeaway on each slide. Or: Create a friendly 8-slide deck teaching basic photo composition to beginners, with short speaker notes.

5. Charts, Diagrams & Visualizations

Visualization work splits into two families. Data charts render numbers: bar and column charts, line and area charts, donut charts, scatter plots, heatmaps, radar charts, histograms, box plots, and waterfall charts. Structural diagrams render relationships and process: flowcharts, mind maps, organization charts, architecture and deployment diagrams, sequence diagrams, entity-relationship diagrams, Gantt timelines, and network or relationship graphs. Both families are delivered as image files (PNG at print resolution) or, for diagrams, as Mermaid-style code you can paste into documentation tools and keep editable.

Chart quality follows publication standards rather than spreadsheet defaults: strong color is reserved for the data itself, gridlines stay light and dashed, top and right borders are removed, legends sit outside the plot area, and labels are checked so they never collide with each other or with the data. Pie charts are drawn as donuts by default, which reads more cleanly at small sizes. For diagrams, connectors are routed so they never pass through boxes, and phase titles are visually distinct from step boxes. If a diagram would need more than about a dozen nodes, it is decomposed: a simple overview diagram carries the big picture and a companion table carries the precise details.

When you need to...	Typical choice	Output
Compare values across categories	Horizontal bar chart	PNG image
Show a trend over time	Smooth line or area chart	PNG image
Show parts of a whole	Donut chart with side legend	PNG image
Explain a process step by step	Flowchart, optionally with swimlanes	PNG or Mermaid code
Brainstorm or organize ideas	Mind map or tree diagram	PNG or Mermaid code
Show how system components connect	Architecture diagram	PNG image
Track project tasks over dates	Gantt chart	PNG image

Table: Choosing the right visual

6. Data Processing & Analysis

Data processing covers everything between having raw files and having an answer. Typical jobs include cleaning (deduplication, missing-value handling, format standardization), transformation (filtering, joining tables, reshaping wide data to long), aggregation (grouping and summarizing), and statistical analysis (distributions, correlations, trends, and comparisons). The work is done in Python with pandas and related libraries, and the scripts themselves can be delivered alongside the results so the process is transparent, repeatable, and easy to rerun when next month's data arrives.

Results arrive in whatever form you ask for: a processed .xlsx or .csv, a set of charts, or a written analysis document that embeds the key numbers and figures with interpretation. For analysis requests, the most useful extra instruction is the question you actually want answered - which channel has the best retention, rather than analyze retention - because that difference determines which summaries and charts get built. Vague input does not stop the work, but a sharp question sharpens the output.

Example request	What you receive
Merge these twelve monthly export files and report totals per region.	A combined dataset plus a summary table, and optionally a chart and a rerunnable script.
Find anomalies in this server log and explain them.	A flagged dataset with an explanation of each anomaly pattern in plain language.
Convert this JSON export into a flat Excel table I can filter.	A formatted .xlsx with typed columns, ready for filtering and pivoting.

Table: Data requests and resulting deliverables

7. Interactive Web Applications

When a document is not enough - when you want something clickable, filterable, or shareable as a live link - the request becomes a web application. Built with Next.js, React, Tailwind CSS, and shadcn/ui components, with optional databases and API routes, this area covers data dashboards with interactive charts and filters, internal tools such as inventory trackers or lightweight CRM systems, calculators and converters, quiz and form apps, and marketing or product landing pages. The result is a runnable application with a real interface, not a mockup.

Choose a web app over a document when the content changes over time, when users need to interact with it by sorting, filtering, editing, or logging in, or when the same view must serve many records. Choose a document when the content is fixed and meant to be printed, signed, or archived. Hybrid needs are

common and worth naming explicitly: for example, a dashboard for daily use plus a PDF export button for monthly reporting gives you both the live view and the permanent record.

Example request	What you receive
Build a personal finance dashboard that tracks spending by category.	A runnable web app with charts, category filters, and editable entries.
Make a simple booking page for my piano studio with a request form.	A landing page with a working form, confirmation state, and a clean mobile layout.
Create a team retro board where we add cards and vote.	An interactive app with shared state, voting, and sorting by score.

Table: Web app requests and resulting deliverables

8. AI Media Services

Media services produce audio and visual assets directly. Image generation turns a text description into an original image - blog illustrations, product concept art, poster backgrounds, character sketches - across a range of aspect ratios and styles. Image editing takes an existing image and modifies it according to instructions: restyle it, extend it, remove or replace elements, or produce variations. Text-to-speech converts text into natural-sounding narration audio with selectable voices and adjustable speed. Speech-to-text transcribes audio recordings into text. Video understanding analyzes an uploaded video and answers questions about its content, motion, and sequence. Image search retrieves real photographs from the web for a described subject, returned with source URLs.

Service	Input	Output	Example request
Image generation	Text description	Image file	A minimalist flat illustration of a lighthouse at dawn, muted blue palette.
Image editing	Image + instructions	Edited image	Replace this product photo's background with clean studio gray.
Text-to-speech	Text	Audio file	Narrate this 300-word intro with a calm voice at normal speed.
Speech-to-text	Audio file	Transcript	Transcribe this interview recording and mark speaker changes.
Video understanding	Video + question	Written analysis	Summarize what happens in this 2-minute demo video, in order.
Image search	Subject description	Web images + URLs	Find five photos of terraced rice fields for my travel blog.

Table: Media services, inputs, and outputs

Two habits improve media results noticeably. For images, describe the style, mood, and palette you want - minimalist flat illustration versus photorealistic, muted versus vivid - and name the intended use so the composition fits the layout. For audio, specify the voice character and pace, and mark any words that must be pronounced exactly as written. If the first render misses the mark, a one-line correction such as warmer lighting, less text in frame usually lands it on the second try.

9. Research & Information Retrieval

Research tasks combine web search with full-page reading. Search finds current information beyond built-in knowledge - recent news, prices, regulations, product releases, event results - and page extraction pulls the complete content of a specific URL into clean text for quoting, summarizing, or archiving. Combined, they support competitive landscape summaries, fact-finding with sources, weekly what-changed briefings, and reading lists assembled around a topic. A typical research answer arrives as a structured summary with the source links listed alongside, so you can verify or read deeper.

Two honest limits are worth planning around. First, source quality varies across the web; for high-stakes claims, ask for the originating sources rather than secondary write-ups, and verify anything critical yourself - links are never invented, but a found source can still be a weak one. Second, very fresh information can be indexed incompletely; if a search returns thin results, narrow the question, name the organizations likely to publish on it, or share documents you already hold so the analysis can proceed on solid ground.

Example request	What you receive
Find the latest 2026 regulations on short-term rentals in Barcelona and summarize the key rules.	A structured summary of the rules with dated sources and official links.
Read this article URL and extract the three main arguments.	A clean extraction with the arguments quoted and briefly assessed.
Compare the top five project management tools for a 10-person team.	A comparison table across price, features, and fit, with links to each vendor.

Table: Research requests and resulting deliverables

10. File & Document Processing

File processing covers manipulation of existing documents - often the fastest wins, because nothing needs to be written from scratch. PDF work includes merging several files into one, splitting by page ranges, extracting text, tables, or embedded images, removing blank pages, rotating or cropping pages, adding watermarks, compressing oversized files, encrypting and decrypting, filling forms, and editing metadata such as title and author. Conversion jobs move between formats: Word, Excel, PowerPoint, HTML, and images to PDF; PDF pages back to PNG images; and scanned pages made searchable through OCR.

Operation	What it does
Merge / split	Combine multiple PDFs into one document, or split one file into chapters or single pages.
Extract	Pull raw text, structured tables, or embedded images out of a PDF for reuse.
OCR	Turn scanned or photographed pages into searchable, selectable text.
Convert	Office files, HTML, or images to PDF; PDF pages back to images.
Watermark / stamp	Overlay text or an image on every page, behind or above the content.
Compress / encrypt	Reduce file size for sharing, or add password protection and permissions.
Forms & metadata	Read and fill PDF form fields; set title, author, and subject metadata.

Table: Common file operations

Batch requests are welcome and usually quick: convert every Word file in a folder to PDF, extract the second page from each of forty scans, or rename and re-organize a directory of mixed documents by type

and date. When a job touches many files, the original files are left untouched and the outputs are written to a separate location, so nothing is lost if a conversion needs to be redone differently.

11. How to Phrase Effective Requests

The single biggest predictor of a good first draft is the specificity of the request. A complete request names six things: the deliverable and format, the audience, the length or size, the tone or style, the must-include facts or assets, and any constraints such as required sections, things to avoid, or how the result will be used. None of this needs to be formal - a sentence or two covering most of the list is enough - but each missing item is a guess the assistant has to make, and guesses are where rework comes from.

Vague request	Sharpened version
Make a presentation about our company.	Make a 10-slide PPTX introducing our company to first-time investors: what we do, market size, traction, team, and the ask. Clean business style, one takeaway line per slide.
Analyze my data.	Analyze orders.csv: monthly revenue trend for 2025, top 10 products by margin, and anomalies in the returns column. Output a short PDF with charts plus the cleaned file.
Write something for my blog.	Write an 800-word blog post for beginner gardeners on starting a balcony herb garden, friendly and practical, ending with a 6-item checklist.

Table: Vague requests versus sharpened ones

The six-part request

Deliverable + audience + length + style + must-includes + constraints. Cover most of the six and the first draft lands close to final.

Iteration is expected and cheap, so treat the first deliverable as a proposal about direction. Ask for structural changes first - sections, ordering, scope - then style changes such as tone, density, or palette, and only then line-level edits. When something is right, say so explicitly, for example keep this structure, so later revisions do not drift away from what already worked. Referencing pages, sections, or table numbers when giving feedback makes each revision pass faster and more precise.

12. Workflows, Best Practices & FAQ

Workflows for larger projects

Larger requests benefit from a light workflow rather than one giant instruction. For a multi-part project - a report plus a deck plus a one-page summary, for instance - agree on the outline first, produce the longest deliverable next, and derive the shorter pieces from it so the messaging stays consistent across formats. Provide materials early in the process: your own data, brand colors, prior documents worth matching, and examples of outputs you like are all absorbed directly into the result. For recurring jobs, such as a monthly report, keep the instruction that worked and reuse it verbatim next cycle; only the data changes.

Quality checks you get for free

Every document-class deliverable passes automated checks before delivery: metadata completeness, font embedding and glyph coverage, table fit against the page width, table-of-contents integrity with clickable links, blank-page removal, and a full-page layout scan for overflow or void issues. These checks catch most problems before you ever see the file. If something still looks off to you, describe what and where - page number, section, or table - and the fix is usually a single pass rather than a rebuild.

Frequently asked questions

Can you work from my existing files? Yes. Attach or point to the file and say what to do with it: revise, extend, convert, extract, or use it as a style reference. Existing documents are often the best possible input because they carry your voice and structure with them.

What formats can be delivered? The common set is PDF, Word, Excel, PowerPoint, PNG and SVG images, MP3 and WAV audio, HTML, CSV and JSON data, and runnable web applications. If the target format is unusual, ask - conversion between many text and data formats is straightforward.

Can you match a style I already have? Yes. Share the document, brand colors, or even a screenshot of a layout you like, and the deliverable will follow that reference for typography, palette, and density instead of a generic look.

How long can documents be? Anywhere from a one-page memo to a full-length handbook with chapters, tables, and charts. Length is your call; as a reference point, the option you choose typically ranges from short documents of a few pages to deep documents of ten or more.

Why did my request come back as questions instead of a file? Clarifying questions fire when key dimensions of the deliverable are genuinely ambiguous - usually audience, length, or format. Answering them takes one message and prevents the most common kind of rework. If you would rather skip the questions, say just do it and reasonable defaults will be applied.

What should I not share? Apply the same judgment you would with any cloud service: avoid sharing credentials, secrets, and personal data that a task does not strictly need. Anonymizing a dataset before handing it over is quick and removes most privacy concerns while changing nothing about the analysis.